ProveIt research submission

Reciprocal pooling and integral projections

Two explicit Diophantine equations for the syntactic consistency of ZFC

A quartic in 218 integer unknowns
A degree-ten companion in 215 integer unknowns

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AI-assisted mathematical research, 25 October 2026

Result and scope. This submission gives two complete, fixed integer-coefficient polynomials, with literal residual lists, expanded coefficients, ordinary mathematical proofs, and independent exact checkers. Composing the new reductions with the retained direct C222/refutation calibration proves in ZFC that each polynomial has no integer zero if and only if Con(ZFC).
The new arithmetic idea merges power certificates at *different* bases through integrality of a rational sum of squared errors. A common sign and a strict size bound replace separate divisibility tests. The quartic combines the newest same-base parent with seven exact integer coordinate eliminations. No global minimum, literature-wide priority, proof of consistency, or new end-to-end proof-assistant certificate is claimed. The polynomial data are complete; the logical calibration is an identified inherited theorem.

Abstract

Starting from the repository's direct 225-integer quartic, we construct a quartic in 218 unrestricted integer variables and 155 quadratic residuals. Its expanded polynomial has 34,984 nonzero monomials and coefficient height 113,304. Five common-Hessian directions and two affine relations provide exact integral projections; a protected change of residual basis lowers coefficient height. The larger support is an explicit cost of this reduction.

We then prove a distinct-base reciprocal-pooling theorem. Fourteen power requests at the consecutive bases q and $q+1$ can share one integer quotient, replacing four quadratic residuals and four private coordinates by one quintic residual and one coordinate. The resulting single equation has 215 unrestricted integer variables, exact degree ten, 152 residuals, 99,914 collected monomials, and the same coefficient height 113,304. Its equivalence with the parent is proved in both directions, including negative indices, signed Pell parameters, unit bases, and the bootstrap that makes both bases nonzero. No coprimality between the two moduli is required. Both polynomials encode the direct numerical proof calculus, without K215, Jones's universal polynomial, or Sun's reduction.

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1 The equations and the exact claim

1.1 A completely specified quartic

All 218 coordinates are independent unrestricted integers. Define

$$\boxed{P_{218}(x_0,\dots,x_{217}) = \sum_{i=0}^{154} Q_i(x_0,\dots,x_{217})^2 = 0.} \tag{1}$$

Appendix A prints every coefficient of every Q_i. Each Q_i has degree at most two. The file `equations/quartic218/equation.json` gives the same list in an exact sparse format; its companion `expanded_polynomial.json` gives all 34,984 collected terms of the single quartic. Thus the summation in (1) is a finite polynomial abbreviation, not a variable-length computation or an unspecified universal-polynomial index.

For the degree-ten companion, let $v = (v_0,\dots,v_{214}) \in \mathbb{Z}^{215}$ and put

$$\widehat{Q}_i(v) = Q_i(v_0,\dots,v_{213},0,0,0,0), \qquad q = v_{95} - 1, \quad A = v_{135} + 1.$$

Define the following polynomial abbreviations:

$$T_0 = 2Aq - q^2 - 1, \qquad T_1 = 2A(q+1) - (q+1)^2 - 1, \tag{2}$$

$$E_0 = \widehat{Q}_{154} + q\widehat{Q}_{153}, \qquad E_1 = \widehat{Q}_{152} + (q+1)\widehat{Q}_{151}, \tag{3}$$

$$\mathcal{R} = E_0T_1 + E_1T_0 - v_{214}T_0T_1. \tag{4}$$

The second equation is

$$\boxed{P_{215}(v_0,\dots,v_{214}) = \sum_{i=0}^{150} \widehat{Q}_i(v)^2 + \mathcal{R}(v)^2 = 0.} \tag{5}$$

Equations (2)–(5), together with the literal appendix, specify the second polynomial completely. Its coefficient files are under `equations/decic215/`. There is no division, exponent variable, factorial, binomial primitive, or hidden coefficient parameter in either final polynomial.

Theorem 1.1 (Consistency equivalence)**.** *For the effective ZFC axiom predicate and finite proof system used in the retained C222 calibration,*

$$\mathrm{ZFC} \vdash \big(\exists x \in \mathbb{Z}^{218}\ P_{218}(x) = 0\big) \iff \exists p\ \mathrm{Ref}_{\mathrm{ZFC}}(p),$$
$$\mathrm{ZFC} \vdash \big(\exists v \in \mathbb{Z}^{215}\ P_{215}(v) = 0\big) \iff \exists p\ \mathrm{Ref}_{\mathrm{ZFC}}(p).$$

Consequently, ZFC proves the absence of integer solutions of either equation equivalent to $\mathrm{Con}(\mathrm{ZFC})$.

Meaning of the proof claim. The new reductions below are ordinary mathematical proofs formalizable in ZFC. Their composition uses the inherited ordinary arithmetic proof of the 225-variable parent and the named finite syntactic translations. The archive preserves those sources and interfaces. It does not contain a newly checked, complete Lean/Rocq derivation of Theorem 1.1, or a literal encoded ZFC proof object. This is a distinction about the form of verification, not an additional assumption that ZFC is consistent.

1.2 Measured comparison

Equation	Variables	Residuals	Degree	Terms	Height
Pinned parent P_{225}	225	162	4	7,867	128,446
Radical projection P_{220}	220	157	4	7,790	142,238
Principal P_{218}	**218**	**155**	**4**	**34,984**	**113,304**
Companion P_{215}	**215**	**152**	**10**	**99,914**	**113,304**

Table 1: All domains are unrestricted integers. Terms and height refer to the fully expanded, collected single polynomial.

Relative to its actual 225-variable parent, the principal saves seven variables and seven residuals, retains degree four, and lowers height by 15,142, about 11.8%. Its support increases by 27,117 monomials. The 215-variable companion saves another three variables and three residuals at the cost of degree ten and a further 64,930 monomials. The maximum absolute coefficient in an individual residual is 240 in both new equations, versus 256 in the parent.

These are improvements in the direct, modest-coefficient C222 branch. They are not absolute fewest-variable records. The inspected repository already has a 37-integer quartic on a different coefficient-heavy branch, and low-variable high-degree equations; its maintained display uses 15 positive integers, degree 157, and 66 integer-leaf binary operations. Those measures describe different compromises [3, 4]. We do not replace a Pareto comparison by a single unsupported claim of "the simplest equation."

2 The source and the logical connection

2.1 Pinned source

The comparison snapshot is ProveIt commit

`953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4`.

The immediate mathematical parent is package `158-pooled-powers`, whose principal residual file has SHA-256

`fee84c408828aba3e7ceebe9a37f58509327063aa7fd078fd49c80adf3af4e56`.

That package supplies the 225-variable direct quartic and a complete ordinary proof of its arithmetic interpretation [1]. The generic quadratic-projection ideas and discovery utilities in package 159 supply a second input [2]. Applying those ideas to the new 225-variable parent is a new instance; merely repeating a projection lemma is not presented as a novel theorem.

The source route is

$$\begin{aligned} \text{integer polynomial zero} &\longleftrightarrow \text{finite C222 refutation} \\ &\longleftrightarrow \text{finite ZF refutation} \\ &\longleftrightarrow \text{finite ZFC refutation.} \end{aligned}$$

The last two bridges are the retained interfaces

```
KEndpoint.c222_iff_zf_refutation
ConstructibleChoice.zf_refutation_iff_zfc_refutation.
```

The source proof also retains its square-shell syntax recoding and signed-Pell representation. All these are dependencies of the result; the present work does not claim to have independently rebuilt the full transitive formal development.

2.2 What the arithmetic source certifies

The C222 route encodes a finite derivation directly. A sufficiently large finite horizon, factorial weights, and CRT residues encode the local proof-rule tuples and the evolution of pending formulas. Exact powers provide binomial digit and range certificates. The signed Pell framework enables all final variables to range over $\mathbb{Z}$ while recovering the semantic signs needed by the proof interpretation.

For the change made here, the crucial retained consequences are precise:

(i) There is an integer W with $W = 2 + \sum_{j=1}^{36} a_j^2$ for the listed affine frame atoms. The horizon atom $t = 16 + t_0^2$ implies $W \geq 258$.

(ii) Before the two pooled arithmetic blocks are decoded, the common Pell parameter satisfies $|A| > W^W$ and the retained Pell pairs have their canonical signed indices.

(iii) Every actual base, requested output, and requested index of the fourteen pooled requests has square strictly below W.

(iv) Three singleton requests are retained: $(F^2+1)^t = V$, $B^x = N$, and $(BH)^x = q$. The normalized ceiling/remainder argument yields $|B| \geq 2$; the multiplier H is odd and nonzero.

The parent proves these facts in an orientation-first, noncircular order. Its proof of the common frame is not replaced by a numerical experiment. The exact changes of coordinates below transport the entire residual system, including these rows and their witnesses.

2.3 Why this is an ordinary Diophantine equation

The powers W^W and n^k appear in the *proof* of the arithmetic interpretation. They are not operations in the final polynomial. The finite quadratic residuals encode the corresponding witnesses, and (5) adds only a quintic polynomial. This separation is essential: a short exponential-Diophantine formula would not by itself answer the question posed here. The classical reduction literature also distinguishes the number of unknowns, degree, and witness domain rather than treating them as interchangeable measures [6, 7].

3 Seven integral eliminations at degree four

3.1 Common-Hessian directions

Lemma 3.1 (Quadratic projection along a common radical)**.** *Let $R_0, \dots, R_{m-1} \in \mathbb{Z}[x_0, \dots, x_{n-1}]$ have degree at most two. Suppose $d \in \mathbb{Z}^n$ lies in the kernel of every Hessian, $d_p = 1$, and for some row r the integer $a_r = \sum_j d_j[x_j]R_r$ is 1 or -1. Put $a_i = \sum_j d_j[x_j]R_i$. For $y_p = 0$, define*

$$Q_i(y) = R_i(y) - a_i a_r R_r(y) \quad (i \neq r).$$

Then the old and new integer zero sets are in explicit bijection. One variable and one residual disappear, and all remaining residuals are quadratic.

Proof. The common-Hessian condition gives the exact polynomial identity $R_i(y + sd) = R_i(y) + sa_i$. Set $s = -a_r R_r(y)$; this makes row r zero and gives $R_i(y + sd) = Q_i(y)$ for every other row. Conversely, for an old zero x, put $y = x - dx_p$. Then $y_p = 0$ and the pivot equation gives $x_p = -a_r R_r(y)$. These two integral maps are inverse on the zero sets. No division by a nonunit occurs. The formula for Q_i is a constant row combination, so its degree stays at most two. □

Five applications give the 220-variable intermediate. The following indices refer to the input at each step; they are not all indices in the original 225-variable system.

Dimension	Pivot p	Deleted row r	a_r	Direction d
225	44	29	1	$e_{44} - e_{45}$
224	60	46	-1	$e_{60} - e_{61} - 2e_{62}$
223	64	49	1	$e_{64} - e_{62}$
222	31	16	1	$e_{31} - e_{32} - 2e_{33}$
221	25	10	1	$e_{25} - e_{27} - 2e_{28}$

The frozen certificates include the complete quadratic restoration polynomial at each step. The independent checker substitutes those maps into every residual and checks both inverse identities. It does not infer equivalence from matching dimensions or sampled evaluations.

3.2 An integral chart of an affine hyperplane

In the 220-variable system, the difference of rows 19 and 6 is

$$4 + 12x_9 + 16x_{10} + 4x_{20} - x_{21} + x_{34} = 0. \tag{6}$$

A direct substitution can make coefficients large. Instead use the following integral parametrization, with four new coordinates u, v, w, z:

$$x_9 = v + 2w, \qquad x_{10} = -v - w, \qquad x_{20} = -1 + v - 2w + z,$$
$$x_{21} = u + 2z, \qquad x_{34} = u - 2z.$$

It satisfies (6) identically. Every integer point on the hyperplane has the integral inverse

$$v = -x_9 - 2x_{10}, \qquad w = x_9 + x_{10},$$
$$z = 1 + 3x_9 + 4x_{10} + x_{20}, \qquad u = x_{21} - 2z.$$

The equation of the hyperplane ensures the restored x_{34} is correct. Substitute the chart in every row and omit row 6. Row 19 remains, so their known difference restores the omitted row. This gives 219 variables and 156 residuals. The maps are explicit; no numerical lattice-reduction heuristic is a premise of the proof.

3.3 The final affine elimination

In the resulting 219-variable chart, the surviving affine row is

$$0 = 4 - 4y_{10} - y_{20} - 4y_{21} - 2(y_{23} + y_{27} + y_{29} + y_{31} + y_{33}) \\ - 4y_{35} - 2(y_{36} + y_{39} + y_{42}) - 4y_{61} - 2y_{62} + y_{94}.$$

Its coefficient at y_{94} is one. Solve for y_{94}, substitute in every other row, and delete that row and coordinate. The inverse inserts the displayed affine expression. This gives 218 variables and 155 quadratic residuals over exactly the same integer domain.

3.4 A protected change of residual basis

The raw affine projection has a large coefficient height. We apply 669 literal operations of the form $R_i \leftarrow R_i + cR_j$, $c \in \mathbb{Z}$, $i \neq j$. Each is invertible by subtraction, so the common zero set is unchanged. The certificate lists every operation in order and the independent checker also replays the inverses in reverse order.

The last four rows, 151–154, are protected: they are neither changed nor used as source rows in this selected basis transformation. This preserves the two old power pools literally, which simplifies the next reduction. The resulting sum of squares is the principal (1). The polynomial itself changes when the residual basis changes; the integer zero set is preserved. The improved height therefore must be measured from the newly expanded polynomial, as it is in Table 1.

4 Reciprocal pooling across distinct bases

4.1 The canonical Pell congruence

For an integer A with $|A| \geq 2$, define the signed canonical Pell coordinates by

$$(X_k, Y_k) = (A, 1)^k \quad (k \in \mathbb{Z})$$

in the group with multiplication

$$(X, Y)(X', Y') = (XX' + (A^2 - 1)YY',\ XY' + YX').$$

The identity is $(1, 0)$ and $(A, 1)^{-1} = (A, -1)$. Thus every coordinate is an integer and $X_k^2 - (A^2-1)Y_k^2 = 1$.

Let $n \neq 0$ and set $T = 2An - n^2 - 1$. Modulo T, $(n - A)^2 \equiv A^2 - 1$. Therefore the map $(X, Y) \mapsto X + (n - A)Y$ is multiplicative and sends $(A, 1)$ to n. Also $T \equiv -1 \pmod{n}$, so n is invertible modulo T. It follows for all signed indices that

$$X_k + (n - A)Y_k \equiv n^k \pmod{T}. \tag{7}$$

Negative powers in this congruence mean powers of the modular inverse. They will be cleared before any size estimate is used.

4.2 Statement of the new compression lemma

Theorem 4.1 (Two-base reciprocal pooling)**.** *Let $W \in \mathbb{Z}$, $W \geq 258$, $A \in \mathbb{Z}$, and $|A| > W^W$. Let $n_0 = q$, $n_1 = q + 1$ be nonzero integers with $n_0^2 < W$ and $n_1^2 < W$. Partition at most 15 requests into two sets I_0, I_1. For each request $i \in I_j$, let $m_i, k_i \in \mathbb{Z}$ satisfy*

$$n_j^2 < W, \qquad m_i^2 < W, \qquad k_i^2 < W,$$

and let $(X_i, Y_i) = (X_{k_i}(A), Y_{k_i}(A))$ be its canonical signed Pell coordinates. Set $Z_i = AY_i$, and define

$$\begin{aligned} T_j &= 2An_j - n_j^2 - 1, \\ H_j &= \sum_{i \in I_j} Y_i(m_i - X_i), \\ S_j &= \sum_{i \in I_j} [(m_i - X_i + Z_i)^2 - Y_i^2], \\ E_j &= S_j - 2n_j H_j. \end{aligned}$$

Then the following are equivalent:

(a) *Every request is the exact power $m_i = n_j^{k_i}$; when $|n_j| \geq 2$, every k_i is nonnegative. For a unit base the exact signed unit power is meant, without an index-sign conclusion.*

(b) *There exists $z \in \mathbb{Z}$ such that*

$$\boxed{E_0T_1 + E_1T_0 - zT_0T_1 = 0.} \tag{8}$$

The moduli T_0, T_1 need not be coprime.

4.3 Proof: integrality, a common sign, and a strict bound

Put $b = \lfloor \sqrt{W-1} \rfloor$. Then $b \geq 16$ and every bounded base, output, and index has absolute value at most b. First,

$$|T_j| \geq 2|A||n_j| - n_j^2 - 1 \geq 2|A| - W > W^W. \tag{9}$$

The sign of T_j is the sign of An_j. Nonzero consecutive integers have the same sign, so T_0 and T_1 have the same sign.

Write $\delta_i = m_i - X_i + Z_i - n_jY_i$. Direct expansion gives

$$\sum_{i \in I_j} \delta_i^2 = E_j - T_j \sum_{i \in I_j} Y_i^2. \tag{10}$$

Consequently (8) says that

$$\sum_{j=0}^{1} \frac{\sum_{i \in I_j} \delta_i^2}{T_j} \in \mathbb{Z}. \tag{11}$$

The quantities δ_i may be very large. Bounding them directly would be incorrect. Instead define the nonzero integer

$$C = (n_0 n_1)^b$$

and the *small cleared power errors*

$$d_i = Cm_i - \frac{C}{n_j^b}\, n_j^{b+k_i}. \tag{12}$$

This is an integer: $b + k_i \geq 0$ and C/n_j^b is an integer. The Pell congruence (7) gives $C\delta_i \equiv d_i \pmod{T_j}$. Therefore

$$\sum_{j=0}^{1} \frac{\sum_{i \in I_j} d_i^2}{T_j} \in \mathbb{Z}. \tag{13}$$

Indeed, multiply (11) by C^2 and replace each numerator by a congruent one modulo its own denominator. The difference is an integer. This argument does not divide by C modulo either modulus and uses no coprimality between the moduli.

The uniform estimates are

$$|C| \leq b^{2b}, \qquad |C/n_j^b| \leq b^b, \qquad |d_i| \leq b^{2b+1} + b^{3b} \leq 2b^{3b}.$$

Writing $r = |I_0| + |I_1| \leq 15$, we obtain

$$\sum_i d_i^2 \leq 4rb^{6b} < W^{3b+1} \leq W^W. \tag{14}$$

The strict inequality uses $4r \leq 60 < W$ and $b^2 < W$. The last uses $3b + 1 \leq b^2 + 1 \leq W$, valid for $b \geq 3$. Together with (9), this shows that the integer in (13) has absolute value strictly less than one. It is zero. All its summands have a common sign, so every d_i is zero.

Divide the equality $d_i = 0$ by C in $\mathbb{Q}$. This gives $m_i = n_j^{k_i}$. If $|n_j| \geq 2$ and $k_i < 0$, then $m_i n_j^{-k_i} = 1$, impossible for an integer m_i. This proves (b) implies (a), including negative bases and negative Pell parameters.

Conversely, true powers and (7) give $T_j \mid \delta_i$. Equation (10) then gives $T_j \mid E_j$. Thus $z = E_0/T_0 + E_1/T_1$ is an integer satisfying (8). This proves (a) implies (b). □

Remark 4.2 (Why both hypotheses matter). Common sign prevents cancellation: $1/2 + 1/(-2) = 0$ has nonzero numerators. A strict bound prevents a nonzero integer sum: $1/2 + 1/2 = 1$. These are counterexamples to weakened rational arguments, not to the theorem. Neither pairwise coprimality nor individual divisibility can simply be read off a rational sum in the absence of the size and sign argument.

4.4 Replacing the old four rows

The old same-base certificates use two private coordinates K_j, L_j per base and the rows

$$A_j = K_j - 2H_j - (2A - n_j)L_j, \tag{15}$$
$$B_j = S_j - n_j K_j + L_j. \tag{16}$$

The exact polynomial identity is

$$B_j + n_j A_j = E_j - T_j L_j. \tag{17}$$

Old zero witnesses give the new quotient $z = L_0 + L_1$. Conversely, the new theorem gives true powers and hence individual integrality of $L_j = E_j/T_j$. Restore $K_j = 2H_j + (2A - n_j)L_j$. Both old rows then vanish.

This is an equivalence after existentially quantifying private witnesses. The inverse quotients are integral *on solutions*; they are not polynomial functions on the full ambient lattice. The report therefore does not assert an ambient polynomial automorphism for this semantic compression.

5 Instantiation and the bootstrap

5.1 The four protected rows

In the principal 218-variable chart, the last four coordinates are

$$x_{214} = K_1, \quad x_{215} = L_1, \quad x_{216} = K_0, \quad x_{217} = L_0.$$

They occur only in rows 151–154. Rows 151,152 are the aggregate and norm for $q + 1$; rows 153,154 are those for q. The actual affine ports are $q = x_{95} - 1$ and $A = x_{135} + 1$. The protected row transformation does not alter any of these four rows.

Setting the four private coordinates to zero in (17) gives exactly (3). The parent has three requests at q and eleven at $q + 1$, hence $r = 14$. Its Pell difference coordinates are exact integral changes of coordinates, and the norm-row simplifications are identities modulo retained Pell rows. The inherited pool certificate and the exact projection maps retain that correspondence. The finite builder does not guess port names or ignore constant offsets: it extracts and verifies their full affine polynomials from the actual four rows.

5.2 Avoiding circular sign reasoning

To apply Theorem 4.1 one must establish $q \neq 0, -1$ before using either pool. The retained singleton requests and the ceiling argument supply $|B| \geq 2$, H odd and nonzero, $x \geq 0$, and $q = (BH)^x$. Therefore $q = 1$ if $x = 0$; otherwise $|q| \geq 2$. In either case $q \neq 0, -1$, so q and $q + 1$ are nonzero.

The theorem now recovers all fourteen requests simultaneously. It is legitimate even in the possible unit-base case $q = 1$. The recovered request $M = q^x$ and the unchanged equation $Bd = M$ rule out $x = 0$: that would imply $B \mid 1$ despite $|B| \geq 2$. Thus $x \geq 1$ and $|q| \geq 2$. The retained integer ratio $q^t v = q^x$ forces $x \geq t$. At this stage the parent's rounded-factorial, CRT range, and Horn-flow arguments resume with exactly their required hypotheses.

For completeness, start with the same finite C222 refutation as in the parent. Choose a sufficiently large cofinal horizon, construct its true power outputs and canonical Pell pairs, and use the parent's witness construction. The new quotient is $L_0 + L_1$. The coordinate projection maps then give a solution of (5). No consistency assumption is used in this conditional witness construction.

5.3 Degree and dimension

Deleting four private variables and adjoining one leaves 215 variables. Deleting four rows and adjoining one leaves 152 residuals. Each E_j is at most cubic and each T_j quadratic, so $\mathcal{R}$ is at most quintic. The coefficient of

$$v_{214}^2 v_{95}^8$$

in the fully expanded (5) is exactly one. Thus degree ten is exact, not merely an upper bound. The other 151 squared residuals have degree at most four and cannot cancel that term.

Proof of Theorem 1.1. The five applications of Lemma 3.1, the affine chart, and the final affine substitution give explicit integer zero-set bijections between P_{225} and the raw 218-variable system. The protected invertible row operations preserve its common zero set, proving the claim for (1). Theorem 4.1, the preceding bootstrap, and the two witness directions after (17) prove that (5) is solvable exactly when (1) is. Compose with the retained parent arithmetic theorem and the C222–ZF–ZFC refutation translations. All steps concern finite syntactic objects, integer arithmetic, recurrences, and finite constructions, and formalize in ZFC. Negation gives the stated consistency equivalences. □

6 What was verified

6.1 Independent exact checks

The principal checker uses only Python's standard library. It imports no construction or optimization module. Its checks are symbolic integer coefficient identities, not searches for solutions. They cover:

- every common-radical condition, complete lift and inverse map, and every projected residual;
- the affine row consequence, the integral hyperplane chart, its inverse, and the final unit-coefficient substitution;
- all 669 residual row operations and their exact inverses;
- privacy of the four removed coordinates, literal preservation of the four pool rows, and every pool elimination/recombination identity;

- all coefficients of both expanded final polynomials, exact degrees, coefficient heights, term counts, and occurrence of every coordinate;
- deliberate corruptions of maps, source coefficients, moduli, row indices, and reindexing data.

The separate arithmetic checker verifies the Pell congruence and rational identities on finite examples with both signs and negative indices. Its 396 signed-congruence cases include 180 negative indices. Eight true-power and eight perturbed-output cases use the full theorem hypotheses with $W = 258$ and $|A| = 258^{258} + 1$. The proof of the unbounded theorem is Section 4, not an extrapolation from these fixtures.

The immediate 225-variable parent's verifier was replayed with a rebuild of its 20 generated JSON stage files. The restored source closure was authenticated against its retained hashes and original pinned archives. The normal and optimized Python checks of the new work agree; acceptance does not rely on `assert`, which optimized Python would remove. Detailed receipts and the exact verification scope are in `verification/`.

6.2 Reproducibility and evaluation

From the extracted archive directory, with Python 3.10 or later:

```
python3 -B verify.py
python3 -B -O verify.py
python3 -B verify.py --rebuild
python3 -B code/evaluate.py quartic218 point.json
python3 -B code/evaluate.py decic215 point.json
```

An evaluator input is exactly one JSON list of 218 or 215 integers, respectively. It evaluates that specified tuple. It does not decide whether a zero exists. The verifier works offline and creates no network requests. The report can be rebuilt by running pdfLaTeX twice in `report/`; the generated residual and metric inputs are included.

6.3 Proof boundaries

An exact coefficient audit does not prove the general Pell or logical calibration theorem. Conversely, a correct general theorem does not authenticate an accidentally mistyped coefficient list. The archive provides both the ordinary proofs and the separate finite checks. The independent reviews are separately assigned AI reviews, not external human peer review. The transitive Lean/Rocq closure was not freshly compiled in this iteration, and no theorem of consistency itself was proved.

7 Novelty, comparison, and the useful next target

The old same-base pooling of package 158 uses divisibility of a sum of squared errors by *one* modulus [1]. The present replacement uses integrality of a rational sum with *two different* moduli. Clearing signed indices, proving a common sign, and budgeting the two denominators against one frame are the extra steps that make this work. The existing reciprocal-factorial CRT construction uses moduli $1 + F^2/i$ and a binomial product identity [5]; it is a different mechanism from reciprocal pooling of Pell-error fractions. We found no instance of the new two-base pooling theorem in the examined snapshot. That is a scoped comparison, not a claim to have proved historical priority over all mathematical literature.

The seven quartic eliminations combine methods already present in the repository with a newer source. Their value is the complete, checked new endpoint and its explicit inverse maps. The affine chart avoids treating rational substitution as if it automatically preserved integer solutions. The protected residual basis makes the semantic power-pool interface stable while changing the surrounding quartic.

The generic principle is reusable: a finite sum of nonnegative rational errors with denominators of one sign vanishes when it is an integer and its absolute value is below one. Here the subtlety is that the actual Pell coordinates are not small; the signed-index clearing and congruence argument replace them by small errors before applying that principle. This suggests a concrete further compiler target: merge additional certificate blocks only after proving a compatible sign orientation and a common cleared-error budget. The present archive claims the completed two-base instance, not a completed arbitrary-base compiler.

For a next reduction, the most valuable question is whether the same cross-base information can be realized with a lower-degree residual without spending back the three saved variables. A stronger global "simplest" claim would require a declared complexity order and a comparison across the coefficient-heavy and positive-domain branches. Neither an undecidability theorem nor the absence of further reductions in one linear-algebra search supplies such a lower bound.

A Every quadratic residual of the principal equation

All variables below are $x_0, \dots, x_{217} \in \mathbb{Z}$. The polynomial is the sum of the squares of exactly these 155 rows. Indices are zero-based both here and in the JSON. In particular the last four rows are the protected two-base interface used in (3).

$$
\begin{aligned}
Q_0 &= -x_2^2 + x_3 \\
Q_1 &= x_0 - x_0x_5 - x_5 + x_8 \\
Q_2 &= -x_2 - x_5x_6 + x_7x_8 \\
Q_3 &= -1 + x_1 - x_7x_8 \\
Q_4 &= 1 + x_3 - x_4x_{19} + x_{17} \\
Q_5 &= -1 - x_1 + x_7x_8 + x_{10} - x_{20}^2 + x_{21} \\
Q_6 &= -4x_9^2 - 4x_9x_{10} - 8x_9x_{20} - 16x_9x_{21} - 2x_{10} - x_{10}x_{20} - 8x_{10}x_{21} \\
&\quad - x_{11} - x_{11}^2 + x_{11}x_{20} + 3x_{20} + x_{20}^2 - 16x_{20}x_{21} + 5x_{21} + x_{22} + x_{72} \\
Q_7 &= -2 - 15x_0 - 5x_0x_5 - x_1 + x_3 - x_3x_{140} - x_4x_{19} - 3x_5 + x_5x_{17} \\
&\quad + x_7x_8 + 5x_8 - 240x_9 - 56x_9^2 + 8x_9x_{10} + 16x_9x_{20} - 128x_9x_{21} \\
&\quad - 64x_9x_{22} + 87x_{10} + 44x_{10}^2 + 42x_{10}x_{11} + 71x_{10}x_{20} - 32x_{10}x_{21} \\
&\quad - 24x_{10}x_{22} + 56x_{10}x_{24} - 8x_{10}x_{25} + 8x_{10}x_{35} - 8x_{10}x_{53} + 22x_{11} \\
&\quad + 14x_{11}^2 - 6x_{11}x_{12} - 3x_{11}x_{20} - 28x_{11}x_{21} + 16x_{11}x_{24} - 16x_{11}x_{53} \\
&\quad + 2x_{12} - 2x_{12}^2 - 50x_{20} - 96x_{20}^2 + 32x_{20}x_{21} - 48x_{20}x_{22} - 8x_{20}x_{25} \\
&\quad + 8x_{20}x_{35} - 16x_{20}x_{53} - 188x_{21} - 60x_{21}^2 - 128x_{21}x_{22} + 16x_{21}x_{24} \\
&\quad - 16x_{21}x_{25} - 51x_{22} - 4x_{22}^2 + 11x_{23} + 16x_{24} - 16x_{24}^2 + 19x_{25} + 2x_{26} \\
&\quad - 8x_{26}^2 + 12x_{27} + 10x_{29} - 2x_{30} - 4x_{30}^2 + 11x_{31} - 2x_{32} + 10x_{33} + 3x_{34} \\
&\quad + 20x_{35} - 8x_{35}^2 + 12x_{36} + 5x_{37} + 10x_{39} + 3x_{40} + 10x_{42} - 8x_{53} - 8x_{53}^2 \\
&\quad + 2x_{54} - 5x_{60} + 12x_{61} - 8x_{61}^2 + 12x_{62} + 9x_{72} - x_{72}^2 + x_{73} - x_{91} \\
&\quad - x_{135}x_{176} - x_{139} - x_{140} + x_{142}x_{143} - x_{176} + x_{177} + 20x_{185}x_{198} \\
&\quad + 20x_{188}^2 - 3x_{192} + 2x_{192}x_{193} - 2x_{200}
\end{aligned}
$$

$$
\begin{aligned}
Q_{8} = {} & -x_{10}-x_{10}^{2}+2x_{10}x_{11}-2x_{10}x_{20}-x_{11}^{2}+2x_{11}x_{20}-x_{21}+x_{24}\\
Q_{9} = {} & -1-x_{10}-3x_{11}^{2}+2x_{11}x_{12}+x_{12}^{2}+2x_{20}-x_{21}+x_{24}+x_{25}-x_{40}\\
& +2x_{72}\\
Q_{10} = {} & -x_{0}-3x_{9}+32x_{9}x_{21}-2x_{10}+11x_{10}^{2}+14x_{10}x_{11}-9x_{10}x_{20}\\
& -8x_{10}x_{21}-4x_{10}x_{24}-24x_{10}x_{25}+16x_{10}x_{44}+8x_{10}x_{53}-14x_{11}\\
& -14x_{11}^{2}+4x_{11}x_{12}-27x_{11}x_{20}-24x_{11}x_{21}-24x_{11}x_{24}-8x_{11}x_{25}\\
& +20x_{11}x_{28}-4x_{11}x_{44}+16x_{11}x_{53}+2x_{12}^{2}-23x_{20}-6x_{20}^{2}\\
& -16x_{20}x_{21}-16x_{20}x_{24}-48x_{20}x_{25}+16x_{20}x_{44}+16x_{20}x_{53}-2x_{21}\\
& -8x_{21}x_{25}-x_{22}+2x_{24}-8x_{24}^{2}-8x_{24}x_{25}-16x_{24}x_{44}-17x_{25}\\
& +8x_{25}^{2}+16x_{25}x_{28}+x_{26}-4x_{26}^{2}+x_{27}+x_{28}+4x_{28}^{2}+x_{34}+x_{37}\\
& -2x_{40}-x_{43}-6x_{44}-4x_{44}^{2}+x_{45}+8x_{53}+8x_{53}^{2}-2x_{54}-2x_{60}-3x_{72}\\
& +x_{188}^{2}\\
Q_{11} = {} & -x_{0}+x_{10}^{2}+5x_{10}x_{20}+x_{11}+4x_{11}^{2}-2x_{11}x_{12}-x_{11}x_{20}-x_{12}^{2}\\
& +11x_{20}+2x_{20}^{2}-8x_{20}x_{24}+x_{21}+15x_{24}-4x_{24}^{2}-x_{25}-4x_{26}^{2}+x_{27}\\
& +x_{40}-x_{44}-3x_{72}+x_{188}^{2}\\
Q_{12} = {} & -2x_{0}-3x_{0}^{2}-2x_{0}x_{3}-x_{1}^{2}-x_{2}^{2}-x_{3}-2x_{3}^{2}-x_{4}^{2}-2x_{5}^{2}+x_{10}\\
& +x_{10}^{2}-2x_{10}x_{11}+2x_{10}x_{20}+14x_{11}-5x_{11}^{2}+2x_{11}x_{12}-16x_{11}x_{24}\\
& -16x_{11}x_{25}-x_{13}^{2}-x_{14}^{2}-x_{15}^{2}-x_{16}^{2}-x_{17}^{2}-x_{18}^{2}-4x_{20}+2x_{20}^{2}\\
& +x_{21}+4x_{24}-4x_{24}^{2}-8x_{24}x_{25}+5x_{25}-4x_{25}^{2}+x_{28}-x_{40}-4x_{72}\\
& +x_{95}-2x_{95}^{2}-x_{96}^{2}-x_{97}^{2}-x_{101}^{2}-x_{104}^{2}-x_{107}^{2}-x_{110}^{2}-x_{113}^{2}\\
& -x_{114}^{2}-x_{117}^{2}-x_{120}^{2}-x_{121}^{2}-x_{125}^{2}-x_{128}^{2}-x_{131}^{2}+x_{134}+x_{140}^{2}\\
& -x_{146}x_{147}+x_{149}^{2}-2x_{152}+x_{152}^{2}+x_{155}^{2}+x_{158}^{2}+x_{161}^{2}+x_{164}^{2}\\
& +x_{167}^{2}+x_{170}^{2}+x_{173}^{2}+x_{175}+x_{176}^{2}+x_{176}x_{187}+x_{179}^{2}+x_{182}^{2}\\
& -x_{185}^{2}-x_{187}^{2}+x_{205}^{2}+x_{206}^{2}+x_{207}+x_{207}^{2}\\
Q_{13} = {} & 2+x_{1}-x_{3}+x_{4}x_{19}-x_{7}x_{8}-x_{9}-2x_{10}-x_{10}^{2}+4x_{10}x_{11}\\
& +4x_{10}x_{40}+23x_{11}-2x_{11}x_{12}+4x_{11}x_{20}-16x_{11}x_{21}+8x_{11}x_{25}\\
& +4x_{11}x_{28}+8x_{11}x_{40}-x_{12}^{2}-x_{17}-4x_{20}+2x_{20}^{2}+8x_{20}x_{25}\\
& +8x_{20}x_{40}+8x_{21}-16x_{21}x_{25}-8x_{21}x_{28}-4x_{24}+8x_{24}x_{25}+4x_{25}\\
& +4x_{25}^{2}-4x_{26}^{2}+x_{27}+12x_{28}+x_{29}-3x_{40}+4x_{40}^{2}-x_{41}-x_{43}-x_{44}\\
& +4x_{72}\\
Q_{14} = {} & -4x_{0}-x_{0}x_{5}-x_{5}+x_{8}-2x_{9}+4x_{9}^{2}+4x_{9}x_{10}+8x_{9}x_{20}\\
& -16x_{9}x_{21}-9x_{10}-2x_{10}^{2}-4x_{10}x_{11}-4x_{10}x_{20}-8x_{10}x_{24}\\
& +4x_{10}x_{25}+16x_{10}x_{44}+4x_{10}x_{53}+20x_{10}x_{56}-3x_{11}+7x_{11}^{2}\\
& -2x_{11}x_{12}-4x_{11}x_{20}+36x_{11}x_{21}+24x_{11}x_{24}-4x_{11}x_{44}+8x_{11}x_{53}\\
& -x_{12}^{2}-6x_{20}-3x_{20}^{2}+32x_{20}x_{21}-16x_{20}x_{24}+4x_{20}x_{25}+16x_{20}x_{44}\\
& +8x_{20}x_{53}+20x_{20}x_{56}+4x_{21}+8x_{21}x_{25}+9x_{24}-4x_{24}^{2}-16x_{24}x_{44}\\
& -16x_{24}x_{56}-3x_{25}-x_{26}-8x_{26}^{2}+2x_{27}+x_{30}+4x_{30}^{2}-x_{31}-x_{34}\\
& +x_{37}+x_{40}-9x_{44}-4x_{44}^{2}+x_{45}-12x_{49}-4x_{49}^{2}+4x_{53}+4x_{53}^{2}-x_{54}\\
& -4x_{56}^{2}+x_{57}-x_{60}+14x_{72}+5x_{188}^{2}
\end{aligned}
$$

$$\begin{aligned}Q_{15} = {} & 2 + 2x_{0} + 2x_{1} - x_{2} - x_{3} + x_{4}x_{19} - x_{5}x_{6} - x_{7}x_{8} + 2x_{9} \\ & - 32x_{9}x_{21} + 14x_{10} - 17x_{10}^{2} - 2x_{10}x_{11} + 8x_{10}x_{21} + 4x_{10}x_{25} \\ & + 4x_{10}x_{53} + 16x_{10}x_{60} - 28x_{11} - 11x_{11}^{2} + 4x_{11}x_{12} - 10x_{11}x_{20} \\ & - 40x_{11}x_{24} + 8x_{11}x_{25} + 4x_{11}x_{28} + 8x_{11}x_{53} + 4x_{11}x_{58} - 2x_{12} + x_{12}^{2} \\ & - x_{17} - 6x_{20} + 11x_{20}^{2} + 16x_{20}x_{21} + 4x_{20}x_{25} + 8x_{20}x_{53} - 8x_{21}x_{24} \\ & - 8x_{21}x_{25} - 8x_{21}x_{28} + 8x_{21}x_{60} + x_{22} - 2x_{23} + 2x_{24} - 12x_{24}^{2} + 13x_{25} \\ & - x_{26} - 8x_{26}^{2} + 12x_{28} - x_{29} + x_{30} - 2x_{31} + x_{32} - 2x_{33} - x_{34} - 4x_{35} \\ & - 2x_{36} - x_{37} - 2x_{39} - 2x_{40} - 2x_{42} - x_{43} - 2x_{44} + 4x_{53} + 4x_{53}^{2} - x_{54} \\ & + 4x_{58}^{2} - x_{59} - 10x_{60} + 4x_{60}^{2} - 4x_{61} - 2x_{62} - 18x_{72} + x_{72}^{2} - x_{73} \\ & + x_{135}x_{176} + x_{176} - x_{177} - 4x_{185}x_{198} - 2x_{188}^{2} + x_{189}^{2} + x_{190}^{2} + x_{192} \\ & - x_{199}^{2}\end{aligned}$$

$$\begin{aligned}Q_{16} = {} & 1 + x_{0} + x_{1} - x_{3} + x_{4}x_{19} - x_{7}x_{8} + 17x_{10} - 2x_{10}^{2} - 6x_{10}x_{11} \\ & - 7x_{10}x_{20} - 16x_{10}x_{24} + 4x_{11} - 4x_{11}^{2} + 2x_{11}x_{12} + 3x_{11}x_{20} \\ & - 4x_{11}x_{21} + 8x_{11}x_{24} + 16x_{11}x_{25} + x_{12}^{2} - x_{17} - 6x_{20} + 13x_{20}^{2} \\ & + 8x_{20}x_{24} - 4x_{21} - 8x_{21}x_{24} - 2x_{23} + 5x_{24} + 4x_{24}^{2} + 8x_{24}x_{25} - 4x_{25} \\ & + 4x_{25}^{2} + 4x_{26}^{2} - 3x_{27} - x_{28} - 2x_{29} - 2x_{31} + x_{32} - 2x_{33} - 4x_{35} - 2x_{36} \\ & - 2x_{39} - x_{40} - 2x_{42} + x_{44} - x_{60} - 4x_{61} - 2x_{62} + 9x_{72} + x_{135}x_{176} + x_{176} \\ & - x_{177} - 4x_{185}x_{198} - x_{188}^{2} + x_{189}^{2} + x_{190}^{2} + x_{192} - x_{199}^{2}\end{aligned}$$

$$\begin{aligned}Q_{17} = {} & 1 + 4x_{0} + 3x_{0}^{2} + 2x_{0}x_{3} + x_{0}x_{5} + x_{1} + x_{1}^{2} + x_{3} + 2x_{3}^{2} + x_{4}^{2} \\ & + x_{4}x_{19} + x_{5} + 2x_{5}^{2} - x_{7}x_{8} - x_{8} - 2x_{9} - 6x_{10} + 2x_{10}^{2} - 28x_{10}x_{11} \\ & - 29x_{10}x_{20} - 8x_{10}x_{24} + 8x_{10}x_{25} - 32x_{10}x_{32} - 4x_{10}x_{35} - 4x_{10}x_{40} \\ & + 16x_{10}x_{44} + 4x_{10}x_{53} - 18x_{11} + 9x_{11}^{2} - 2x_{11}x_{12} + 13x_{11}x_{20} \\ & + 4x_{11}x_{21} - 32x_{11}x_{24} + 8x_{11}x_{25} - 40x_{11}x_{28} - 40x_{11}x_{32} - 8x_{11}x_{40} \\ & - 4x_{11}x_{44} + 8x_{11}x_{53} + x_{13}^{2} + x_{14}^{2} + x_{15}^{2} + x_{16}^{2} - x_{17} + x_{17}^{2} + x_{18}^{2} \\ & - 7x_{20} + 5x_{20}^{2} - 40x_{20}x_{24} + 16x_{20}x_{25} - 64x_{20}x_{32} - 4x_{20}x_{35} \\ & - 8x_{20}x_{40} + 16x_{20}x_{44} + 8x_{20}x_{53} + 12x_{21} - 8x_{21}x_{24} - 13x_{24} - 32x_{24}^{2} \\ & + 8x_{24}x_{25} - 16x_{24}x_{32} - 16x_{24}x_{44} + 11x_{25} - 16x_{25}^{2} - 32x_{25}x_{28} \\ & - 8x_{25}x_{32} + 12x_{26}^{2} - 3x_{27} - 4x_{28} - 8x_{28}^{2} - 2x_{30} - 4x_{30}^{2} + x_{31} - 3x_{32} \\ & - 4x_{32}^{2} + x_{33} + 4x_{35}^{2} - x_{36} + x_{37} + 5x_{40} - 4x_{40}^{2} + x_{41} + 2x_{43} + 8x_{44} \\ & - 4x_{44}^{2} + x_{45} + 4x_{53} + 4x_{53}^{2} - x_{54} - 3x_{60} + 35x_{72} - x_{95} + 2x_{95}^{2} + x_{96}^{2} \\ & + x_{97}^{2} + x_{101}^{2} + x_{104}^{2} + x_{107}^{2} + x_{110}^{2} + x_{113}^{2} + x_{114}^{2} + x_{117}^{2} + x_{120}^{2} \\ & + x_{121}^{2} + x_{125}^{2} + x_{128}^{2} + x_{131}^{2} - x_{134} - x_{140}^{2} + x_{146}x_{147} - x_{149}^{2} \\ & + 2x_{152} - x_{152}^{2} - x_{155}^{2} - x_{158}^{2} - x_{161}^{2} - x_{164}^{2} - x_{167}^{2} - x_{170}^{2} - x_{173}^{2} \\ & - x_{175} - x_{176}^{2} - x_{176}x_{187} - x_{179}^{2} - x_{182}^{2} + x_{185}^{2} + x_{187}^{2} - 3x_{188}^{2} \\ & - x_{205}^{2} - x_{206}^{2} - x_{207} - x_{207}^{2}\end{aligned}$$

$$Q_{18} = 1 + x_{9} - 4x_{21}^{2}$$

$$\begin{aligned}Q_{19} = {} & - 2 - x_{1} + x_{7}x_{8} - 8x_{9}^{2} - 8x_{9}x_{10} - 16x_{9}x_{20} + x_{10}^{2} - 2x_{10}x_{11} \\ & - 2x_{11} + 3x_{11}^{2} + 4x_{20} - 2x_{21} + x_{22} - 2x_{24} - x_{25} + x_{34}\end{aligned}$$

$$
\begin{aligned}
Q_{20} = {} & 2 + 12x_0 - 3x_0^2 - 2x_0x_3 + 5x_0x_5 + x_1 - x_1^2 - x_2^2 - x_3 - 2x_3^2 \\
& + x_3x_{140} - x_4^2 + 3x_5 - 2x_5^2 - x_5x_{17} - x_7x_8 - 5x_8 - 190x_9 \\
& - 56x_9^2 + 8x_9x_{10} + 16x_9x_{20} + 64x_9x_{21} - 64x_9x_{34} - 68x_{10} - 86x_{10}^2 \\
& - 32x_{10}x_{11} - 27x_{10}x_{20} + 104x_{10}x_{21} - 36x_{10}x_{24} + 24x_{10}x_{25} \\
& + 32x_{10}x_{32} - 24x_{10}x_{34} - 4x_{10}x_{35} - 4x_{10}x_{40} - 4x_{10}x_{45} - 4x_{10}x_{48} \\
& + 8x_{10}x_{53} + 16x_{10}x_{60} - 15x_{11} - 45x_{11}^2 + 18x_{11}x_{12} - 5x_{11}x_{20} \\
& + 28x_{11}x_{21} - 16x_{11}x_{24} + 24x_{11}x_{25} + 16x_{11}x_{28} + 40x_{11}x_{32} \\
& - 8x_{11}x_{40} + 8x_{11}x_{45} + 8x_{11}x_{48} + 16x_{11}x_{53} + 4x_{11}x_{58} - 4x_{12} \\
& + 6x_{12}^2 - x_{13}^2 - x_{14}^2 - x_{15}^2 - x_{16}^2 + x_{17} - x_{17}^2 - x_{18}^2 + 36x_{20} \\
& + 80x_{20}^2 - 48x_{20}x_{21} + 24x_{20}x_{24} + 24x_{20}x_{25} + 64x_{20}x_{32} - 48x_{20}x_{34} \\
& - 4x_{20}x_{35} - 8x_{20}x_{40} - 4x_{20}x_{45} - 4x_{20}x_{48} + 16x_{20}x_{53} + 150x_{21} \\
& - 48x_{21}^2 - 16x_{21}x_{24} + 40x_{21}x_{25} + 8x_{21}x_{28} + 128x_{21}x_{34} + 8x_{21}x_{60} \\
& + 5x_{22} - 10x_{23} - 28x_{24} + 36x_{24}^2 + 16x_{24}x_{32} + 8x_{25} + 8x_{25}^2 \\
& + 16x_{25}x_{28} + 8x_{25}x_{32} - 3x_{26} - 16x_{26}^2 - 6x_{27} - 11x_{28} + 4x_{28}^2 - 11x_{29} \\
& + 4x_{30} + 4x_{30}^2 - 11x_{31} + 6x_{32} + 4x_{32}^2 - 11x_{33} - 53x_{34} - 4x_{34}^2 - 19x_{35} \\
& + 4x_{35}^2 - 11x_{36} - 13x_{37} - 10x_{39} - 5x_{40} - 4x_{40}^2 + x_{41} - 10x_{42} - 16x_{44} \\
& + 4x_{45}^2 - x_{46} + 4x_{48}^2 - x_{49} + 8x_{53} + 8x_{53}^2 - 2x_{54} + 4x_{58}^2 - x_{59} \\
& + 5x_{60} + 4x_{60}^2 - 12x_{61} + 8x_{61}^2 - 12x_{62} - 77x_{72} + 2x_{72}^2 - 2x_{73} + x_{91} \\
& + 2x_{95} - 2x_{95}^2 - x_{96}^2 - x_{97}^2 - x_{101}^2 - x_{104}^2 - x_{107}^2 - x_{110}^2 - x_{113}^2 \\
& - x_{114}^2 - x_{117}^2 - x_{120}^2 - x_{121}^2 - x_{125}^2 - x_{128}^2 - x_{131}^2 + x_{134} - x_{135} \\
& + x_{135}x_{152} + x_{139} + x_{140} + x_{140}^2 - x_{142}x_{143} + x_{149}^2 - x_{152} + x_{152}^2 \\
& - x_{153} + x_{155}^2 + x_{158}^2 + x_{161}^2 + x_{164}^2 + x_{167}^2 + x_{170}^2 + x_{173}^2 + x_{176}^2 \\
& + x_{179}^2 + x_{182}^2 - x_{185}^2 - 20x_{185}x_{198} - x_{187}^2 - 19x_{188}^2 + x_{189}^2 + x_{190}^2 \\
& + 3x_{192} - 2x_{192}x_{193} - x_{199}^2 + 2x_{200} + x_{205}^2 + x_{206}^2 + x_{207} + x_{207}^2
\end{aligned}
$$

$$
Q_{21} = -x_{10}^2 - 2x_{10}x_{20} + 4x_{10}x_{35} - x_{20}^2 + 4x_{20}x_{35} - 4x_{35}^2 + x_{36}
$$

$$
\begin{aligned}
Q_{22} = {} & 1 - 2x_9 + 15x_{10}^2 + 14x_{10}x_{11} - 4x_{10}x_{20} - 4x_{11} + x_{11}^2 - 4x_{11}x_{20} \\
& + 8x_{11}x_{21} + 4x_{20} + 2x_{20}^2 + 4x_{21} + x_{25} + x_{37} - 2x_{60}
\end{aligned}
$$

$$
\begin{aligned}
Q_{23} = {} & 2x_0 - 14x_{10} - 3x_{10}^2 + 2x_{10}x_{11} + 4x_{10}x_{20} + 8x_{10}x_{24} + 32x_{10}x_{37} \\
& - 16x_{10}x_{44} + 4x_{11} + 5x_{11}^2 - 2x_{11}x_{12} + 4x_{11}x_{20} - 16x_{11}x_{24} \\
& - 24x_{11}x_{37} + 4x_{11}x_{44} - x_{12}^2 + 6x_{20} - 5x_{20}^2 - 8x_{20}x_{24} + 32x_{20}x_{37} \\
& - 16x_{20}x_{44} + 8x_{21} - x_{24} - 8x_{24}x_{37} + 16x_{24}x_{44} - 8x_{26}^2 + 2x_{27} + 4x_{37} \\
& - 4x_{37}^2 + x_{38} + x_{40} - x_{44} + 4x_{44}^2 - x_{45} + 2x_{72} - 2x_{188}^2
\end{aligned}
$$

$$
\begin{aligned}
Q_{24} = {} & 7x_0 + 3x_0^2 + 2x_0x_3 + 2x_0x_5 + x_1^2 + x_2^2 + x_3 + 2x_3^2 + x_4^2 \\
& + 2x_5 + 2x_5^2 - 2x_8 + 9x_{10} + 7x_{10}^2 - 12x_{10}x_{11} - 6x_{10}x_{20} - 4x_{10}x_{24} \\
& - 12x_{10}x_{25} - 32x_{10}x_{38} - 16x_{10}x_{44} + 4x_{10}x_{45} - 21x_{11} + 9x_{11}^2 \\
& - 4x_{11}x_{12} + 12x_{11}x_{20} - 4x_{11}x_{21} - 32x_{11}x_{24} + 8x_{11}x_{25} - 24x_{11}x_{38} \\
& + 4x_{11}x_{44} - 8x_{11}x_{45} - x_{12}^2 + x_{13}^2 + x_{14}^2 + x_{15}^2 + x_{16}^2 + x_{17}^2 \\
& + x_{18}^2 + 12x_{20} - 20x_{20}x_{25} - 64x_{20}x_{38} - 16x_{20}x_{44} + 4x_{20}x_{45} - 10x_{21} \\
& - 8x_{21}x_{24} + 7x_{24} - 8x_{24}^2 + 16x_{24}x_{44} - 3x_{25} - 8x_{25}x_{38} - 4x_{26}^2 + x_{27} \\
& - x_{30} - 4x_{30}^2 + x_{31} + x_{32} + 4x_{38} - 4x_{38}^2 + x_{39} + 2x_{40} + 2x_{44} + 4x_{44}^2 \\
& - x_{45} - 4x_{45}^2 + x_{46} - x_{60} + 6x_{72} - 2x_{95} + 2x_{95}^2 + x_{96}^2 + x_{97}^2 + x_{101}^2 \\
& + x_{104}^2 + x_{107}^2 + x_{110}^2 + x_{113}^2 + x_{114}^2 + x_{117}^2 + x_{120}^2 + x_{121}^2 + x_{125}^2 \\
& + x_{128}^2 + x_{131}^2 - x_{134} + x_{135} - x_{135}x_{152} - x_{140}^2 - x_{149}^2 + x_{152} - x_{152}^2 \\
& + x_{153} - x_{155}^2 - x_{158}^2 - x_{161}^2 - x_{164}^2 - x_{167}^2 - x_{170}^2 - x_{173}^2 - x_{176}^2 \\
& - x_{179}^2 - x_{182}^2 + x_{185}^2 + x_{187}^2 - 7x_{188}^2 - x_{205}^2 - x_{206}^2 - x_{207} - x_{207}^2
\end{aligned}
$$

$$
Q_{25} = -x_{11}^2 - 2x_{11}x_{12} - x_{12}^2 + x_{40}
$$

$$
\begin{aligned}
Q_{26} = {} & 1 - x_0 + x_1 - x_7x_8 + x_{10}^2 + 2x_{10}x_{20} - 4x_{10}x_{40} - 2x_{11} + 4x_{11}^2 \\
& + 2x_{11}x_{20} - 8x_{11}x_{40} + 2x_{20} - 8x_{20}x_{25} - 8x_{20}x_{40} + 2x_{21} + 6x_{24} \\
& - 8x_{24}x_{25} + 20x_{25} - 4x_{25}^2 + 4x_{26}^2 - x_{27} + 4x_{40} - 4x_{40}^2 + x_{41} + x_{44} \\
& - 2x_{72} + x_{188}^2
\end{aligned}
$$

$$
\begin{aligned}
Q_{27} = {} & -2 + 2x_0 - x_1 + x_7x_8 - 32x_9x_{21} - 7x_{10} + 2x_{10}x_{20} + 8x_{10}x_{21} \\
& + 12x_{10}x_{24} - 8x_{10}x_{25} - 28x_{10}x_{40} - 16x_{10}x_{41} - 8x_{10}x_{53} + 25x_{11} \\
& + 6x_{11}^2 - 6x_{11}x_{12} + 10x_{11}x_{20} + 16x_{11}x_{21} + 8x_{11}x_{25} + 4x_{11}x_{28} \\
& - 24x_{11}x_{40} - 16x_{11}x_{41} - 16x_{11}x_{53} + 2x_{12} - 2x_{12}^2 + 14x_{20} + 5x_{20}^2 \\
& + 16x_{20}x_{21} + 8x_{20}x_{24} + 8x_{20}x_{25} - 56x_{20}x_{40} - 32x_{20}x_{41} - 16x_{20}x_{53} \\
& + 10x_{21} - 8x_{21}x_{25} - 8x_{21}x_{28} + x_{22} - 3x_{24} - 8x_{24}^2 + 24x_{24}x_{25} \\
& - 25x_{25} - 32x_{25}x_{40} - 16x_{25}x_{41} - x_{26} - 16x_{26}^2 + 4x_{27} + 12x_{28} + x_{29} \\
& - x_{34} - 5x_{40} - 12x_{40}^2 - 16x_{40}x_{41} - 3x_{41} - 4x_{41}^2 + x_{42} - x_{43} - 2x_{44} \\
& - 8x_{53} - 8x_{53}^2 + 2x_{54} + 12x_{72} - x_{72}^2 + x_{73} - 2x_{188}^2
\end{aligned}
$$

$$
\begin{aligned}
Q_{28} = {} & -1 + x_0 - x_{10} - x_{10}^2 - 2x_{10}x_{11} - 4x_{10}x_{20} - 13x_{11} - 5x_{11}^2 + 2x_{11}x_{12} \\
& - 4x_{11}x_{20} + 16x_{11}x_{24} - 24x_{11}x_{25} - 20x_{11}x_{28} + x_{12}^2 + 2x_{20} - 3x_{21} \\
& - 9x_{24} + 4x_{24}^2 + 8x_{24}x_{25} - 8x_{25} - 12x_{25}^2 - 16x_{25}x_{28} - x_{28} - 4x_{28}^2 \\
& - x_{40} + x_{43} - 2x_{72} - x_{188}^2 - x_{189}^2 - x_{190}^2 + x_{199}^2
\end{aligned}
$$

$$
\begin{aligned}
Q_{29} = {} & -1 - 7x_{10} - x_{10}^2 - 4x_{10}x_{11} - x_{10}x_{20} + 8x_{10}x_{24} - x_{11} + x_{11}^2 \\
& - 3x_{11}x_{20} - 9x_{20} - 2x_{20}^2 + 8x_{20}x_{24} + 2x_{21} + 6x_{24} - 4x_{24}^2 + x_{44} \\
& + x_{72}
\end{aligned}
$$

$$
\begin{aligned}
Q_{30} = {} & -1 + 7x_{10} - 2x_{10}^2 + 2x_{10}x_{11} + 2x_{10}x_{20} + 16x_{10}x_{24} + 16x_{10}x_{44} \\
& - 4x_{10}x_{53} - 8x_{11} + 3x_{11}^2 - 2x_{11}x_{12} + 2x_{11}x_{20} - 8x_{11}x_{24} - 4x_{11}x_{44} \\
& - 8x_{11}x_{53} - x_{12}^2 + 16x_{20} + 6x_{20}^2 + 8x_{20}x_{24} + 16x_{20}x_{44} - 8x_{20}x_{53} \\
& - 2x_{21} - 4x_{24} - 12x_{24}^2 - 16x_{24}x_{44} - 3x_{25} - 4x_{26}^2 + x_{27} + x_{40} - 3x_{44} \\
& - 4x_{44}^2 + x_{45} - 4x_{53} - 4x_{53}^2 + x_{54} - 2x_{72}
\end{aligned}
$$

$$
\begin{aligned}
Q_{31} = {} & 1 + x_{10} + x_{10}^2 + 2x_{10}x_{20} + 4x_{10}x_{45} - 2x_{11}^2 - 8x_{11}x_{45} - x_{20} - x_{20}^2 \\
& + 4x_{20}x_{45} + 2x_{21} - 2x_{24} - 4x_{45}^2 + x_{46}
\end{aligned}
$$

$$Q_{32} = -1 - 4x_{46} - 4x_{46}^2 + x_{47}$$

$$Q_{33} = 1 - x_{11}^2 - 8x_{11}x_{46} - 4x_{11}x_{47} + 8x_{46} - 8x_{46}^2 - 16x_{46}x_{47} - 2x_{47} - 4x_{47}^2 + x_{48}$$

$$Q_{34} = 1 + x_1 - x_7x_8 + x_{10} + x_{10}^2 + 2x_{10}x_{20} + 4x_{10}x_{48} - 2x_{11}^2 - 8x_{11}x_{48} - x_{20} - x_{20}^2 + 4x_{20}x_{48} + 2x_{21} - 2x_{24} - 4x_{48}^2 + x_{49}$$

$$Q_{35} = -4x_{43}^2 - 8x_{43}x_{49} - 4x_{49}^2 + x_{50}$$

$$Q_{36} = -1 - x_1 + x_7x_8 + 3x_{10} + x_{10}^2 + 2x_{10}x_{20} - 8x_{10}x_{43} - 4x_{10}x_{50} - 2x_{11} + 2x_{11}^2 + 2x_{11}x_{20} + 10x_{20} + 3x_{20}^2 - 16x_{20}x_{43} - 8x_{20}x_{50} - 2x_{21} - x_{24} - x_{25} + 24x_{43} - 12x_{43}^2 + 8x_{43}x_{49} - 16x_{43}x_{50} + 4x_{49}^2 + 11x_{50} - 4x_{50}^2 + x_{51}$$

$$Q_{37} = -1 - 4x_{51} - 4x_{51}^2 + x_{52}$$

$$Q_{38} = 1 + x_1 - x_7x_8 - x_{11}^2 - 8x_{11}x_{51} - 4x_{11}x_{52} + 8x_{51} - 8x_{51}^2 - 16x_{51}x_{52} - 2x_{52} - 4x_{52}^2 + x_{53}$$

$$Q_{39} = 1 - 2x_{10} - x_{10}^2 - 2x_{10}x_{11} - 3x_{10}x_{20} - 4x_{10}x_{53} - 5x_{11} - 2x_{11}^2 - 5x_{11}x_{20} - 8x_{11}x_{53} - 5x_{20} - 2x_{20}^2 - 8x_{20}x_{53} - x_{25} - 4x_{53} - 4x_{53}^2 + x_{54} - x_{72}$$

$$Q_{40} = 2 - x_0 + x_1 - x_3 + x_4x_{19} - x_7x_8 + x_{10} - x_{17} - x_{20}^2 + x_{21} - 12x_{54} - 4x_{54}^2 + x_{55} + x_{188}^2$$

$$Q_{41} = 1 + x_1 - x_7x_8 + 4x_{10} + 16x_{10}x_{54} + 8x_{10}x_{55} + 4x_{20} + 16x_{20}x_{54} + 8x_{20}x_{55} - 4x_{24} - 16x_{24}x_{54} - 8x_{24}x_{55} - x_{44} + 8x_{54} - 16x_{54}^2 - 16x_{54}x_{55} + 4x_{55} - 4x_{55}^2 + x_{56}$$

$$Q_{42} = 1 - x_0 + x_1 - x_7x_8 + 3x_{10} - 3x_{10}^2 - 2x_{10}x_{11} - x_{10}x_{20} - 16x_{10}x_{44} + 20x_{10}x_{56} - 5x_{11} - 3x_{11}^2 + 3x_{11}x_{20} + 8x_{11}x_{24} + 4x_{11}x_{44} + 2x_{20}^2 - 16x_{20}x_{44} + 20x_{20}x_{56} + x_{21} - 6x_{24} + 4x_{24}^2 + 16x_{24}x_{44} - 16x_{24}x_{56} - x_{25} - x_{44} + 4x_{44}^2 - x_{45} - 12x_{49} - 4x_{49}^2 - 4x_{56}^2 + x_{57} + 3x_{72} + x_{188}^2 + x_{189}^2 + x_{190}^2 - x_{199}^2$$

$$Q_{43} = 2 + x_0 + x_1 - x_7x_8 + x_{10} - x_{20}^2 + x_{21} + 24x_{49} - 8x_{49}^2 - 16x_{49}x_{57} - 2x_{57} - 4x_{57}^2 + x_{58} - x_{188}^2$$

$$Q_{44} = -1 + x_0 + 7x_{10} + 6x_{10}x_{11} - x_{10}x_{20} - 16x_{10}x_{60} - x_{11} + 2x_{11}^2 - 3x_{11}x_{20} + 4x_{11}x_{21} - 4x_{11}x_{58} + x_{20} + 9x_{20}^2 - 2x_{21} - 8x_{21}x_{60} - x_{24} - 4x_{58}^2 + x_{59} + 11x_{60} - 4x_{60}^2 - x_{72} - x_{188}^2$$

$$Q_{45} = 2 + 2x_1 - x_2 - x_5x_6 - x_7x_8 - 3x_9 + 5x_{10} + 6x_{10}x_{11} - x_{10}x_{20} - 16x_{10}x_{21} - 2x_{11} - x_{11}^2 - 3x_{11}x_{20} + 4x_{11}x_{21} + 3x_{20} + 6x_{20}^2 + 6x_{21} + x_{25} + x_{37} - x_{60} + x_{72}$$

$$Q_{46} = -1 + x_0 - x_1 + x_7x_8 + 2x_9 + 12x_{10} - 14x_{10}^2 + 10x_{10}x_{11} + 3x_{10}x_{20} - 32x_{10}x_{59} - 7x_{11} - 7x_{11}x_{20} + 8x_{11}x_{21} + 8x_{11}x_{58} + 5x_{20} + 8x_{20}^2 - 7x_{21} - 16x_{21}x_{59} - x_{24} + 2x_{25} - x_{37} + 8x_{58}^2 + 14x_{59} - 4x_{59}^2 - 2x_{60} + x_{61} + 3x_{72} - x_{188}^2$$

$$Q_{47} = -1 - x_1 + x_7x_8 - 4x_{61} - 4x_{61}^2 + x_{62}$$

$$Q_{48} = -x_2x_{14} + x_{63}$$
$$Q_{49} = x_{13} - x_{14} - 2x_{23}x_{63} - x_{64}$$
$$Q_{50} = x_{13} - x_{14} - 2x_{27}x_{63} - x_{65}$$
$$Q_{51} = -x_2x_{14} - 2x_{29}x_{63} + 2x_{42}x_{63} - x_{63} - x_{66} + x_{71}$$
$$Q_{52} = x_{13} - x_{14} - 2x_{31}x_{63} - x_{67}$$
$$Q_{53} = x_{13} - x_{14} + 2x_{23}x_{63} - 2x_{29}x_{63} - 2x_{33}x_{63} + x_{64} - x_{66} - x_{68}$$
$$Q_{54} = x_{13} - x_{14} - 4x_{35}x_{63} - 2x_{36}x_{63} - x_{69}$$
$$Q_{55} = x_{13} - x_{14} - 2x_{39}x_{63} - x_{70}$$
$$Q_{56} = -2x_2x_{14} + x_{13} - x_{14} - 2x_{42}x_{63} - 2x_{63} - x_{71}$$
$$Q_{57} = -x_{10} + x_{10}^2 + 3x_{10}x_{20} - x_{11} - x_{11}^2 + x_{11}x_{20} - x_{20} + 2x_{20}^2 + x_{72}$$
$$Q_{58} = 1 + 2x_{12} + x_{12}^2 - x_{72}^2 + x_{73}$$
$$Q_{59} = -x_{13}^2 + 2x_{13}x_{14} + 2x_{13}x_{75} - x_{14}^2 - 2x_{14}x_{75} - x_{73}^2 + x_{74} - x_{75}^2$$
$$Q_{60} = x_2x_{14} - 4x_{61}x_{63} - 2x_{62}x_{63} + x_{63} + x_{75}$$
$$Q_{61} = x_{10}x_{63} - x_{14} + x_{20}x_{63} + x_{76}x_{77} + x_{78}$$
$$Q_{62} = -1 + 3x_2 - 4x_2x_{10} - 2x_2x_{21} - x_5x_6 + x_7x_8 + x_{76}$$
$$Q_{63} = -x_{10}x_{213} - x_{13} - 2x_{20}x_{213} + x_{77} + x_{213}$$
$$Q_{64} = x_2x_{14} - x_{10}x_{213} + x_{13} - x_{14} - x_{20}x_{213} - 2x_{21}x_{63} + x_{63} + x_{79}$$
$$Q_{65} = -x_{13}x_{64} + x_{14}x_{64} + x_{80}$$
$$Q_{66} = -x_{65}x_{80} + x_{81}$$
$$Q_{67} = -x_{66}x_{81} + x_{82}$$
$$Q_{68} = -x_{67}x_{82} + x_{83}$$
$$Q_{69} = -x_{68}x_{83} + x_{84}$$
$$Q_{70} = -x_{69}x_{84} + x_{85}$$
$$Q_{71} = -x_{70}x_{85} + x_{86}$$
$$Q_{72} = -x_{71}x_{86} + x_{87}$$
$$Q_{73} = -x_{78}x_{87} + x_{88}$$
$$Q_{74} = -x_{79}x_{88} + x_{89}$$
$$Q_{75} = -x_{89}^2 + x_{90}$$
$$Q_{76} = -x_5 - x_5x_{17} + x_{17} + x_{91}$$
$$Q_{77} = -x_{15}x_{91} + x_{92}$$
$$Q_{78} = -2x_1^2 + 2x_1x_{13} + 2x_1x_{14} + 2x_1x_{92} + 2x_1x_{93} - x_{13}^2 - 2x_{13}x_{92} - x_{14}^2 - 2x_{14}x_{93} + x_{18}x_{94} - x_{74}x_{90} - x_{92}^2 - x_{93}^2$$
$$Q_{79} = -x_5x_{16} + x_{16} - x_{16}x_{91} + x_{93}$$
$$Q_{80} = -x_{95}x_{98} + x_{98} + x_{100}$$
$$Q_{81} = -x_4x_{121} + x_{96}x_{203} - x_{99} - x_{100}x_{121}$$
$$Q_{82} = x_{101}x_{203} - x_{102}x_{124} - x_{103}$$
$$Q_{83} = x_{104}x_{203} - x_{105}x_{124} - x_{106}$$
$$Q_{84} = x_{107}x_{203} - x_{108}x_{124} - x_{109}$$

$$Q_{85} = x_{110}x_{203} - x_{111}x_{124} - x_{112}$$

$$Q_{86} = x_{113}x_{204} - x_{115}x_{124} - x_{116}$$

$$Q_{87} = x_{117}x_{204} - x_{118}x_{124} - x_{119}$$

$$Q_{88} = -x_{18}x_{121} + x_{124}$$

$$Q_{89} = x_{120} - x_{122}x_{124} - x_{123}$$

$$Q_{90} = -x_{124}x_{126} + x_{125} - x_{127}$$

$$Q_{91} = -x_{124}x_{129} + x_{128}x_{203} - x_{130}$$

$$Q_{92} = -x_{124}x_{132} + x_{131}x_{203} - x_{133}$$

$$Q_{93} = x_{134} - 2x_{134}x_{136} + x_{137}$$

$$Q_{94} = -x_{134}x_{137} - x_{137} + x_{138}$$

$$Q_{95} = x_{135} + x_{135}^2 + x_{137}^2 - x_{138}^2 - x_{189}x_{196} + x_{193}$$

$$Q_{96} = -x_{135}x_{140} - x_{140} + x_{141}$$

$$Q_{97} = -1 + x_{139}^2 + 2x_{139}x_{141} + x_{140}^2$$

$$Q_{98} = x_3^2 - 2x_3x_{135} - x_{135} + x_{142} + x_{189}x_{196} - x_{193}$$

$$Q_{99} = -x_3x_{140} + x_5 - x_{139} - x_{140} + x_{142}x_{143}$$

$$Q_{100} = 1 - 2x_4 + x_4^2 - 2x_4x_{135} + x_{144}$$

$$Q_{101} = -x_4x_{176} + x_{18} + x_{144}x_{145} - x_{175}$$

$$Q_{102} = -1 + 2x_{135}x_{187} - x_{146} + 2x_{187} - x_{187}^2$$

$$Q_{103} = -1 + x_{95} + x_{146}x_{147} - x_{175} - x_{176}x_{187}$$

$$Q_{104} = -x_{135}x_{149} - x_{149} + x_{150}$$

$$Q_{105} = -1 + x_{148}^2 + 2x_{148}x_{150} + x_{149}^2$$

$$Q_{106} = 1 + x_{135} - x_{135}x_{152} - x_{152} + x_{153}$$

$$Q_{107} = x_{151}^2 + 2x_{151}x_{153} - 2x_{152} + x_{152}^2$$

$$Q_{108} = -x_{135}x_{155} - x_{155} + x_{156}$$

$$Q_{109} = -1 + x_{154}^2 + 2x_{154}x_{156} + x_{155}^2$$

$$Q_{110} = -x_{135}x_{158} - x_{158} + x_{159}$$

$$Q_{111} = -1 + x_{157}^2 + 2x_{157}x_{159} + x_{158}^2$$

$$Q_{112} = -x_{135}x_{161} - x_{161} + x_{162}$$

$$Q_{113} = -1 + x_{160}^2 + 2x_{160}x_{162} + x_{161}^2$$

$$Q_{114} = -x_{135}x_{164} - x_{164} + x_{165}$$

$$Q_{115} = -1 + x_{163}^2 + 2x_{163}x_{165} + x_{164}^2$$

$$Q_{116} = -x_{135}x_{167} - x_{167} + x_{168}$$

$$Q_{117} = -1 + x_{166}^2 + 2x_{166}x_{168} + x_{167}^2$$

$$Q_{118} = -x_{135}x_{170} - x_{170} + x_{171}$$

$$Q_{119} = -1 + x_{169}^2 + 2x_{169}x_{171} + x_{170}^2$$

$$Q_{120} = -x_{135}x_{173} - x_{173} + x_{174}$$

$$Q_{121} = -1 + x_{172}^2 + 2x_{172}x_{174} + x_{173}^2$$

$$Q_{122} = 1 - x_{135}x_{176} - x_{176} + x_{177}$$

$$Q_{123} = -1 + 2x_{175} + x_{175}^2 + 2x_{175}x_{177} + x_{176}^2$$

$$Q_{124} = -x_{135}x_{179} - x_{179} + x_{180}$$

$$Q_{125} = -1 + x_{178}^2 + 2x_{178}x_{180} + x_{179}^2$$

$$Q_{126} = -x_{135}x_{182} - x_{182} + x_{183}$$

$$Q_{127} = -1 + x_{181}^2 + 2x_{181}x_{183} + x_{182}^2$$

$$Q_{128} = -x_{135}x_{185} - x_{185} + x_{186}$$

$$Q_{129} = -1 + x_{184}^2 + 2x_{184}x_{186} + x_{185}^2$$

$$\begin{aligned} Q_{130} = {} & -2x_0 - 3x_0^2 - 2x_0x_3 - x_1^2 - x_2^2 - x_3 - 2x_3^2 - x_4^2 - 2x_5^2 - 2x_{11} \\ & + x_{11}^2 + 2x_{11}x_{20} - x_{12}^2 - x_{13}^2 - x_{14}^2 - x_{15}^2 - x_{16}^2 - x_{17}^2 - x_{18}^2 \\ & - x_{24} - x_{25} + x_{95} - 2x_{95}^2 - x_{96}^2 - x_{97}^2 - x_{101}^2 - x_{104}^2 - x_{107}^2 - x_{110}^2 \\ & - x_{113}^2 - x_{114}^2 - x_{117}^2 - x_{120}^2 - x_{121}^2 - x_{125}^2 - x_{128}^2 - x_{131}^2 + x_{134} \\ & + x_{140}^2 - x_{146}x_{147} + x_{149}^2 - 2x_{152} + x_{152}^2 + x_{155}^2 + x_{158}^2 + x_{161}^2 \\ & + x_{164}^2 + x_{167}^2 + x_{170}^2 + x_{173}^2 + x_{175} + x_{176}^2 + x_{176}x_{187} + x_{179}^2 \\ & + x_{182}^2 - x_{185}^2 - x_{187}^2 + x_{205}^2 + x_{206}^2 + x_{207} + x_{207}^2 \end{aligned}$$

$$\begin{aligned} Q_{131} = {} & x_4 - 2x_4x_{11} - 2x_4x_{12} - 2x_4x_{13} - 2x_4x_{14} - 2x_4x_{15} - 2x_4x_{16} \\ & - 2x_4x_{17} - 2x_4x_{20} + x_{187} \end{aligned}$$

$$\begin{aligned} Q_{132} = {} & -2 + x_0 - 2x_1 + x_2 + x_3 - x_4x_{19} + x_5x_6 + x_7x_8 - 2x_{10} - 2x_{10}x_{11} \\ & - x_{10}x_{20} + x_{11} + 2x_{11}^2 - 3x_{11}x_{20} + x_{17} + x_{20} + 2x_{20}^2 - 3x_{21} - x_{24} \\ & - x_{72} - x_{135}x_{176} - x_{176} + x_{177} - x_{188}^2 - x_{189}^2 - x_{190}^2 - x_{191}^2 - x_{192}^2 \\ & + x_{199}^2 + x_{200}^2 \end{aligned}$$

$$Q_{133} = -x_{135}x_{190} - x_{190} + x_{199}$$

$$Q_{134} = -1 + x_{189}^2 + x_{190}^2 - x_{199}^2$$

$$Q_{135} = -x_{192} - x_{192}x_{193} + x_{200}$$

$$Q_{136} = -1 + x_{191}^2 + x_{192}^2 - x_{200}^2$$

$$Q_{137} = -x_{185}^2 + x_{201}$$

$$Q_{138} = x_{190} - 2x_{195}x_{201} - x_{201}$$

$$Q_{139} = x_{135}x_{190} - 3x_{190} - 8x_{190}x_{194} + x_{193} - x_{199}$$

$$Q_{140} = -x_{135} - x_{189}x_{196} + x_{193}$$

$$Q_{141} = -x_{184} - x_{186} - x_{189}x_{197} + x_{191}$$

$$\begin{aligned} Q_{142} = {} & -2x_{10} - x_{20} - 2x_{20}^2 - 2x_{21} - 2x_{23} - 2x_{27} - 2x_{29} - 2x_{31} - 2x_{33} - 4x_{35} \\ & - 2x_{36} - 2x_{39} - 2x_{42} - 4x_{61} - 2x_{62} - 4x_{185}x_{198} + x_{192} \end{aligned}$$

$$
\begin{aligned}
Q_{143} = {} & 2 - x_{0} - x_{0}^{2} - 2x_{0}x_{140} - 2x_{0}x_{149} + x_{1} - 2x_{1}x_{176} - x_{2}^{2} - x_{3} \\
& - x_{3}^{2} + x_{3}x_{140} - 2x_{3}x_{149} - x_{4}^{2} - x_{5} - x_{5}^{2} - 2x_{5}x_{167} - x_{7}x_{8} \\
& + 2x_{10} - 2x_{10}x_{152} + 2x_{10}x_{155} - 8x_{10}x_{185} - 2x_{11}x_{158} - 2x_{12}x_{161} \\
& - 2x_{13}x_{164} - 2x_{14}x_{170} - 2x_{15}x_{173} - 2x_{16}x_{179} - 2x_{17}x_{182} - x_{18}^{2} \\
& + 4x_{20} - 4x_{20}x_{152} + 2x_{20}x_{155} - 2x_{20}x_{185} - 8x_{21}x_{185} - 4x_{23}x_{185} \\
& - 4x_{27}x_{185} - 4x_{29}x_{185} - 4x_{31}x_{185} - 4x_{33}x_{185} - 8x_{35}x_{185} - 4x_{36}x_{185} \\
& - 4x_{39}x_{185} - 4x_{42}x_{185} - 8x_{61}x_{185} - 4x_{62}x_{185} + 2x_{95} - 2x_{95}^{2} - x_{96}^{2} \\
& - x_{97}^{2} - x_{101}^{2} - x_{104}^{2} - x_{107}^{2} - x_{110}^{2} - x_{113}^{2} - x_{114}^{2} - x_{117}^{2} - x_{120}^{2} \\
& - x_{121}^{2} - x_{125}^{2} - x_{128}^{2} - x_{131}^{2} + x_{134} - x_{135}x_{140} - x_{135}x_{149} \\
& - x_{135}x_{155} - x_{135}x_{158} - x_{135}x_{161} - x_{135}x_{164} - x_{135}x_{167} - x_{135}x_{170} \\
& - x_{135}x_{173} - x_{135}x_{179} - x_{135}x_{182} - 3x_{135}x_{185} - x_{135}x_{202} - x_{139} - x_{141} \\
& + x_{141}^{2} - x_{142}x_{143} - 2x_{148} - x_{149} - x_{150} + x_{150}^{2} - 2x_{151} + 2x_{152} - 2x_{153} \\
& + x_{153}^{2} - 2x_{154} - x_{155} - x_{156} + x_{156}^{2} - 2x_{157} - x_{158} - x_{159} + x_{159}^{2} \\
& - 2x_{160} - x_{161} - x_{162} + x_{162}^{2} - 2x_{163} - x_{164} - x_{165} + x_{165}^{2} - 2x_{166} - x_{167} \\
& - x_{168} + x_{168}^{2} - 2x_{169} - x_{170} - x_{171} + x_{171}^{2} - 2x_{172} - x_{173} - x_{174} + x_{174}^{2} \\
& + x_{175}^{2} + 2x_{175}x_{177} + x_{176}^{2} + x_{177}^{2} - 2x_{178} - x_{179} - x_{180} + x_{180}^{2} \\
& - 2x_{181} - x_{182} - x_{183} + x_{183}^{2} - 2x_{184} + 5x_{185} + x_{186} + x_{186}^{2} - x_{187}^{2} \\
& - x_{188}^{2}
\end{aligned}
$$

$$Q_{144} = -x_{97}x_{203} + x_{121}$$

$$Q_{145} = -x_{114}x_{204} + x_{121}$$

$$
\begin{aligned}
Q_{146} = {} & 1 - 5x_{10} + 2x_{10}^2 - 2x_{10}x_{11} - x_{10}x_{20} - 16x_{10}x_{23} - 16x_{10}x_{27} \\
& - 16x_{10}x_{29} - 16x_{10}x_{31} - 16x_{10}x_{33} - 24x_{10}x_{35} - 16x_{10}x_{36} \\
& - 16x_{10}x_{39} - 16x_{10}x_{42} - 32x_{10}x_{61} - 16x_{10}x_{62} - 3x_{11} + 2x_{11}^2 \\
& + x_{11}x_{20} - 3x_{20} - 5x_{20}^2 - 8x_{20}x_{21} - 4x_{20}x_{23} - 4x_{20}x_{27} - 4x_{20}x_{29} \\
& - 4x_{20}x_{31} - 4x_{20}x_{33} - 4x_{20}x_{36} - 4x_{20}x_{39} - 4x_{20}x_{42} - 8x_{20}x_{61} \\
& - 4x_{20}x_{62} - 16x_{21}x_{23} - 16x_{21}x_{27} - 16x_{21}x_{29} - 16x_{21}x_{31} \\
& - 16x_{21}x_{33} - 32x_{21}x_{35} - 16x_{21}x_{36} - 16x_{21}x_{39} - 16x_{21}x_{42} \\
& - 32x_{21}x_{61} - 16x_{21}x_{62} + 4x_{23} - 4x_{23}^2 - 8x_{23}x_{27} - 8x_{23}x_{29} \\
& - 8x_{23}x_{31} - 8x_{23}x_{33} - 16x_{23}x_{35} - 8x_{23}x_{36} - 8x_{23}x_{39} - 8x_{23}x_{42} \\
& - 16x_{23}x_{61} - 8x_{23}x_{62} - 2x_{24} - x_{25} + 4x_{27} - 4x_{27}^2 - 8x_{27}x_{29} \\
& - 8x_{27}x_{31} - 8x_{27}x_{33} - 16x_{27}x_{35} - 8x_{27}x_{36} - 8x_{27}x_{39} - 8x_{27}x_{42} \\
& - 16x_{27}x_{61} - 8x_{27}x_{62} + 4x_{29} - 4x_{29}^2 - 8x_{29}x_{31} - 8x_{29}x_{33} \\
& - 16x_{29}x_{35} - 8x_{29}x_{36} - 8x_{29}x_{39} - 8x_{29}x_{42} - 16x_{29}x_{61} - 8x_{29}x_{62} \\
& + 4x_{31} - 4x_{31}^2 - 8x_{31}x_{33} - 16x_{31}x_{35} - 8x_{31}x_{36} - 8x_{31}x_{39} \\
& - 8x_{31}x_{42} - 16x_{31}x_{61} - 8x_{31}x_{62} + 4x_{33} - 4x_{33}^2 - 16x_{33}x_{35} \\
& - 8x_{33}x_{36} - 8x_{33}x_{39} - 8x_{33}x_{42} - 16x_{33}x_{61} - 8x_{33}x_{62} + 8x_{35} \\
& - 24x_{35}^2 - 16x_{35}x_{36} - 16x_{35}x_{39} - 16x_{35}x_{42} - 32x_{35}x_{61} - 16x_{35}x_{62} \\
& + 6x_{36} - 4x_{36}^2 - 8x_{36}x_{39} - 8x_{36}x_{42} - 16x_{36}x_{61} - 8x_{36}x_{62} - x_{37} \\
& + 4x_{39} - 4x_{39}^2 - 8x_{39}x_{42} - 16x_{39}x_{61} - 8x_{39}x_{62} + 4x_{42} - 4x_{42}^2 \\
& - 16x_{42}x_{61} - 8x_{42}x_{62} + 12x_{61} - 12x_{61}^2 - 16x_{61}x_{62} + 3x_{62} - 4x_{62}^2 \\
& + x_{72} - x_{140}^2 - x_{149}^2 + 2x_{152} - x_{152}^2 - x_{155}^2 - x_{158}^2 - x_{161}^2 - x_{164}^2 \\
& - x_{167}^2 - x_{170}^2 - x_{173}^2 - x_{176}^2 - x_{179}^2 - x_{182}^2 + x_{185}^2 - 24x_{185}x_{198} \\
& + 4x_{192} - 2x_{192}x_{193} + 2x_{200} - x_{205}^2 - x_{206}^2 - x_{207} - x_{207}^2
\end{aligned}
$$

$$
\begin{aligned}
Q_{147} = {} & -1 - 4x_{99}^2 - 4x_{103}^2 - 4x_{106}^2 - 4x_{109}^2 - 4x_{112}^2 - 4x_{116}^2 - 4x_{119}^2 + x_{121}^2 \\
& - 4x_{123}^2 - 4x_{127}^2 - 4x_{130}^2 - 4x_{133}^2 - x_{208}^2 - x_{209}^2 - x_{210} - x_{210}^2 \\
& - 4x_{212}^2
\end{aligned}
$$

$$Q_{148} = -x_4x_{211} + x_{121}$$

$$Q_{149} = x_2x_{121} - x_5x_{211} - x_{212}$$

$$Q_{150} = -x_2x_{13} + x_{213}$$

$$
\begin{aligned}
Q_{151} = {} & x_{95}x_{215} - 2x_{96}x_{149} + 2x_{101} - 2x_{101}x_{152} - 2x_{104}x_{155} - 2x_{107}x_{158} \\
& - 2x_{110}x_{161} - 2x_{113}x_{164} - 2x_{117}x_{170} - 2x_{120}x_{173} - 2x_{125}x_{179} \\
& - 2x_{128}x_{182} - 2x_{131}x_{185} - 2x_{135}x_{215} + 2x_{148}x_{149} + 2x_{149}x_{150} - 2x_{151} \\
& + 2x_{151}x_{152} + 2x_{152}x_{153} - 2x_{153} + 2x_{154}x_{155} + 2x_{155}x_{156} + 2x_{157}x_{158} \\
& + 2x_{158}x_{159} + 2x_{160}x_{161} + 2x_{161}x_{162} + 2x_{163}x_{164} + 2x_{164}x_{165} \\
& + 2x_{169}x_{170} + 2x_{170}x_{171} + 2x_{172}x_{173} + 2x_{173}x_{174} + 2x_{178}x_{179} \\
& + 2x_{179}x_{180} + 2x_{181}x_{182} + 2x_{182}x_{183} + 2x_{184}x_{185} + 2x_{185}x_{186} + x_{214} \\
& - 2x_{215}
\end{aligned}
$$

$$\begin{aligned}
Q_{152} = {} & -1 - x_{95}x_{214} + x_{96}^2 - 2x_{96}x_{148} + x_{101}^2 - 2x_{101}x_{151} + x_{104}^2 \\
& - 2x_{104}x_{154} + x_{107}^2 - 2x_{107}x_{157} + x_{110}^2 - 2x_{110}x_{160} + x_{113}^2 \\
& - 2x_{113}x_{163} + x_{117}^2 - 2x_{117}x_{169} + x_{120}^2 - 2x_{120}x_{172} + x_{125}^2 \\
& - 2x_{125}x_{178} + x_{128}^2 - 2x_{128}x_{181} + x_{131}^2 - 2x_{131}x_{184} + x_{148}^2 - x_{149}^2 \\
& + x_{151}^2 + 2x_{152} - x_{152}^2 + x_{154}^2 - x_{155}^2 + x_{157}^2 - x_{158}^2 + x_{160}^2 - x_{161}^2 \\
& + x_{163}^2 - x_{164}^2 + x_{169}^2 - x_{170}^2 + x_{172}^2 - x_{173}^2 + x_{178}^2 - x_{179}^2 + x_{181}^2 \\
& - x_{182}^2 + x_{184}^2 - x_{185}^2 + x_{215}
\end{aligned}$$

$$\begin{aligned}
Q_{153} = {} & x_{95}x_{217} - 2x_{97}x_{140} - 2x_{114}x_{167} - 2x_{121}x_{176} - 2x_{135}x_{217} \\
& + 2x_{139}x_{140} + 2x_{140}x_{141} + 2x_{166}x_{167} + 2x_{167}x_{168} + 2x_{175}x_{176} + 2x_{176} \\
& + 2x_{176}x_{177} + x_{216} - 3x_{217}
\end{aligned}$$

$$\begin{aligned}
Q_{154} = {} & -x_{95}x_{216} + x_{97}^2 - 2x_{97}x_{139} + x_{114}^2 - 2x_{114}x_{166} + x_{121}^2 - 2x_{121}x_{175} \\
& + x_{139}^2 - x_{140}^2 + x_{166}^2 - x_{167}^2 + x_{175}^2 - x_{176}^2 + x_{216} + x_{217}
\end{aligned}$$

B Exact data conventions and identities

A sparse term `[c,[i,j,...]]` denotes $c\,x_i x_j \cdots$; an empty index list denotes the constant c. Repeated indices are repeated factors. A residual file declares its coordinate order, unrestricted-integer domain, and sum-of-squares combiner. Expanded files list the coefficients of the single collected polynomial itself. SHA-256 identifiers authenticate bytes or the named canonical JSON encoding; they do not replace the polynomial definitions.

Artifact	SHA-256 of the supplied file
quartic218/residuals	`878d2f427a117f35b4c211a223003978`
	`d52509acc46a5c97b829380b6e608ad8`
quartic218/expanded polynomial	`f81c0390e593d503d9274bd20c718385`
	`f8a20b6f7aa3f2fa5b74a43cf0157c34`
decic215/residuals	`a335e0d1d1d82aa2cd30c2f436ed61a7`
	`52fff001db134b339c23fae01d4a70b0`
decic215/expanded polynomial	`3f2c8eadf41cc85b54b5a55ad1cfb4fd`
	`0f5100f1373723c706e836a788737d6e`

The frozen transformation plan starts at the retained original 225-variable residual file, replays five radical projections, the affine hyperplane chart, one affine substitution, and the protected row basis, then emits the reciprocal companion. The final standalone equations can be evaluated without executing a source generator or opening an archive.

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